ubeconn

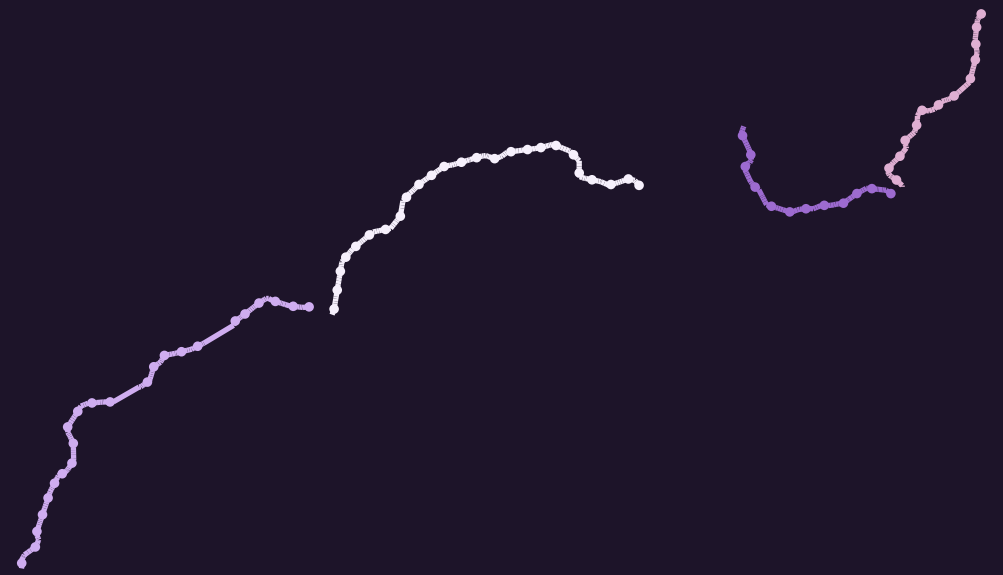
From fibre signals to cable intelligence

Distributed acoustic sensing for cable monitoring • Project status and proposed next steps

SEE THE SIGNAL. UNDERSTAND THE EVENT.

A traceable path from DAS recordings to mapped evidence, expert review and validated alerts.

BUILD > VERIFY > LEARN > OPERATE



LOFOTEN / 4 IMPORTED ROUTES

Current position: a working local investigation platform with a clear path to field validation.

ubeconn

The foundation is built. Field validation is the next gate.

Progress spans capture, asset context, investigation and exploratory analytics.

BUILT

Evidence workspace

Map assets, inspect raw windows, replay recordings and record classification reviews.

DEMONSTRATED

Discovery workflow

A bounded archive pilot extracts features and ranks candidate windows for expert review.

NEXT

Validated detection

Verify timing and geography; collect independent labels; evaluate by incident and season.

579
imported masts

575
inferred connections

115,847
legacy RMS event records

4
target event classes

A strong software foundation; classification accuracy and operational benefit remain to be measured.

SOURCES / docs/map-app.md · docs/masts.md · docs/event-ml.md · recorded DB snapshot

ubeconn

Turn distributed measurements into actionable evidence

The intended operator outcome: find where to investigate, why it matters and what supports the decision.

TARGET CLASS 01

Normal Operations

Establish the range of normal behaviour, including wind.

TARGET CLASS 02

Ice Loading

Test whether signal changes can distinguish ice-related conditions.

TARGET CLASS 03

Wind Galloping

Validate sustained motion patterns against field observations.

TARGET CLASS 04

External Interference

Review strikes, vegetation contact and other disturbances.

MEASURE > LOCATE > REVIEW > PRIORITISE ACTION

Proposed value: earlier investigation and better maintenance prioritisation, measured during the pilot.

SOURCES / AGENTS.md · docs/event-ml.md; operational outcomes are project goals

ubeconn

One evidence chain connects signals, assets and review

HDF5 stores the signal; PostgreSQL stores context; the API serves bounded analysis to the browser.

INPUT

DAS archive

Interrogator
Completed HDF5 files
Explicit channel IDs

>

PROCESSING

Bounded analysis

Python / h5py
Fixed-frame metrics
Replay + file snapshots

>

SERVICE

FastAPI

Signal windows
Asset geometry
Events and review API

>

OPERATOR

Cable Observatory

React / MapLibre
Plotly signal review
Classification history

POSTGRESQL / assets · events · weather · recording references · review history

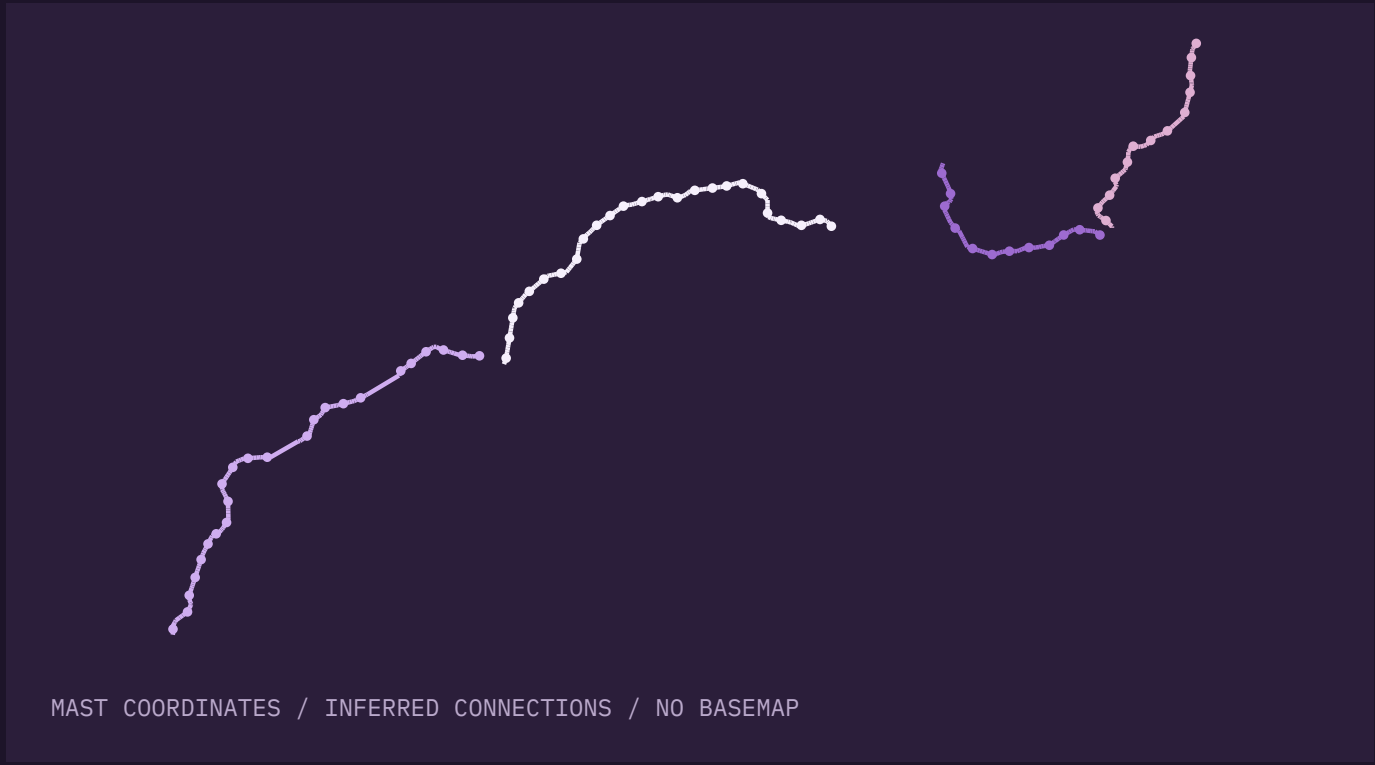
The local investigation workflow is implemented; verified live capture and production rollout are separate gates.

SOURCES / docs/map-app.md · src/Mapui/{api,signals,metrics,live}.py

ubeconn

579 masts anchor the asset view across four routes

Workbook provenance and channel references are retained for every imported mast.



ROUTE	MASTS / CONNECTIONS
R06 Fygle–Solbjørn	188 / 187
R12 Svolvær–Kleppstad	94 / 93
R13 Kvitfossen–Svolvær	101 / 100
R14 Kleppstad–Fygle via Valberg	196 / 195

Assets are mapped; physical connectivity, water crossings and distant endpoint matches still need review.

ubeconn

Investigate an event, a place or a recording

An integrated local workspace connects the event queue, network map and signal evidence.



LOCATE

Search routes, masts and recorded events

INSPECT

Waveform, heatmap, spectrum and trends

REVIEW

Store a class, evidence note and history

ACTUAL LOCAL BUILD / ORIGINAL UI COLOURS

SOURCE / .local/screenshots/overview.png · docs/map-app.md

ubeconn

Signal amplitude and event meaning stay distinct

Map layers are reproducible window measurements. Classification requires independent evidence.

MAP METRIC

AC RMS

Vibration level after removing the mean

MAP METRIC

Band RMS

Energy within a selected frequency band

MAP METRIC

Absolute peak

Largest mean-removed signal excursion

MAP METRIC

Variance

Signal variability in squared units

TRACEABILITY

File + sample window + channel ID + scale + processing version

17,010

exact coordinate matches

8,224

bounded coordinate interpolations

2

channels remain unlocated

Never interpret map colour as event class or severity; no missing fibre ranges are reconstructed.

SOURCES / docs/map-app.md · docs/channel-mapping.md · src/Mapui/metrics.py

ubeconn

The largest archive is still an unlabeled candidate set

Time, location and labels must be verified together before historical events become training evidence.

Legacy alarms	APR–JUN 2025	**115,847 RMS records**	Class and reliable event location unresolved
Channel metadata	APR 2026	**25,235 positions**	Historical mapping equivalence unverified
Example recordings	JUN 2026*	**3 full local files**	Independent of the historical alarm archive
Expert ground truth	NEXT PHASE	**Incident / inspection evidence**	Required to validate the four target classes

* Header date is plausible but not independently verified.

Alarm volume is useful for review prioritisation; it does not establish labeled incidents or model accuracy.

ubeconn

Weather adds context once time and location are trustworthy

YR collection is in place at two configured locations; humidity is a planned extension.

COLLECTED TODAY

Temperature · wind speed · direction

Kråkern Bru and Leknes
YR forecast context
Missing and stale values remain visible

PROPOSED EXTENSION

Humidity + source-validity timestamps

Explicit variable and unit definitions
Collector and storage support
Documented station-to-route relevance

ALIGNMENT RULE

Verified acquisition time > appropriate location > latest valid prior weather

Weather supports interpretation; humidity or freezing conditions alone do not prove ice loading.

ubeconn

Archive discovery has been exercised on real recordings

A reproducible pilot produces a compact review queue without loading the full archive into memory.

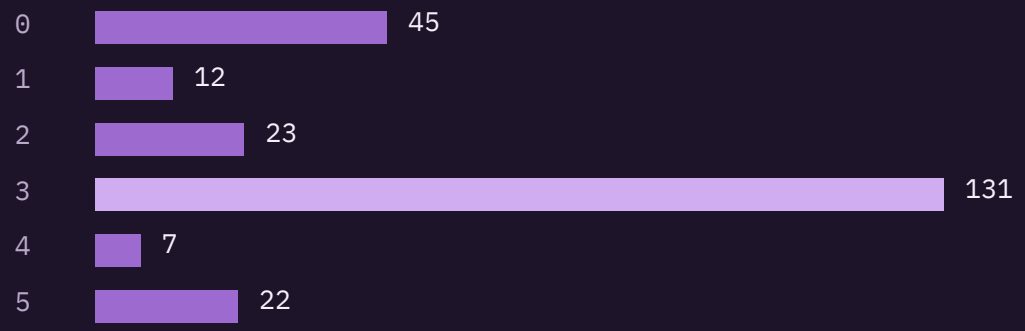
90
archive files

15 min
approximate coverage

240
window / block feature rows

6
exploratory clusters

CLUSTER POPULATION / FEATURE ROWS



RECORDED EXECUTION

37.8 s initial / 0.29 s resume

~160 MiB peak resident memory
Identical feature output after resume
One pilot run; no accuracy estimate

This proves the discovery workflow on a small archive; seasonal coverage and verified labels are still needed.

ubeconn

The ML baseline is built; field accuracy is still unproven

The implementation supports four classes, grouped validation and quantified prediction windows.

01

Discover

Rank unusual and representative windows.

02

Verify

Attach inspection, camera or incident evidence.

03

Train

Use one agreed signal representation.

04

Evaluate

Hold out independent incidents and seasons.

RELEASE EVIDENCE

Per-class precision / recall · false-alert burden · confidence calibration
Site and season generalisation · detection latency · traceable predictions

Model deployment follows field evidence; threshold alarms and anomaly clusters cannot substitute for labels.

ubeconn

Software checks do not establish field performance

Recorded local checks support the investigation workflow; operational acceptance remains ahead.

CAPABILITY	EVIDENCE AVAILABLE	REMAINING GATE
Asset foundation	579 masts / 575 inferred links	Confirm physical inventory
Investigation app	Browser workflow + review persistence	Authenticated multi-user deployment
Signal and replay	Bounded reads / 3 files replayed	Verified live timing and mapping
Discovery	240 feature rows / repeatable resume	Broader seasonal data
Four-class baseline	Training + grouped evaluation code	Labels and held-out field results

RECORDED LOCAL CHECKS: 58 PYTHON TESTS + FRONTEND BUILD + BROWSER REVIEW CHECKS

The next milestone is a controlled field pilot with measurable acceptance criteria.

ubeconn

Advance through evidence gates, then expand deployment

Sequence proposed for planning; durations, owners and budget require project agreement.

GATE 1

Trust the data

Verify acquisition time, channel mapping and preprocessing.

Exit: traceable reference windows

GATE 2

Build ground truth

Collect field labels, normal baselines and richer weather context.

Exit: reviewed, grouped dataset

GATE 3

Prove detection

Evaluate held-out incidents; calibrate thresholds and confidence.

Exit: agreed performance achieved

GATE 4

Pilot operations

Run shadow alerts, measure workload and harden deployment.

Exit: operator acceptance

Protect the evidence chain first; expand sites, seasons and automation only after acceptance gates pass.

ubeconn

Measure detection quality and operational usefulness

Agree numerical targets before the pilot; the project does not yet have validated operating thresholds.

MEASURE	PILOT EVIDENCE	PROPOSED OWNER
Data integrity	Verified timestamps, channel coverage, missing-data rate	Acquisition + data owner
Event quality	Per-class precision / recall; false alerts per route-day	Analytics + domain reviewer
Location and latency	Error against verified location; acquisition-to-alert delay	Field + platform owner
Operational value	Review time, actionable findings, missed confirmed incidents	Operations owner
Robustness	Performance across sites, seasons and configuration changes	Project acceptance team

Pilot success means trusted evidence and a manageable alert burden under representative field conditions.

ubeconn

Prepare the platform for controlled operations

A production pilot adds identity, reliability and incident handling around the existing evidence workflow.

IDENTITY + GOVERNANCE

Accountable decisions

Authenticated reviewers
Restricted database access
Reviewed schema migration
Versioned models and thresholds

RELIABILITY

Observable operation

Capture health and stale-data status
Storage and retention planning
Backup / restore verification
Live / replay separation

INCIDENT WORKFLOW

Useful alerts

Merge related candidate windows
Preserve evidence references
Human review and closure
Agreed escalation integrations

SHADOW MODE FIRST

Observe decisions and workload before automated operational escalation.

Deploy a support tool with traceable evidence; increase automation as field confidence grows.

SOURCES / docs/map-app.md deployment boundary · docs/event-ml.md next steps; extensions proposed

ubeconn

Fund the next evidence milestone

The proposed next increment is a controlled validation pilot with named owners and a reviewed dataset.

DECISION 1

Choose the pilot scope

Representative routes, field access and independently observable conditions.

DECISION 2

Assign the evidence owners

Acquisition and mapping, expert labeling, analytics and operator review.

DECISION 3

Agree the acceptance gate

Detection quality, alert workload, traceability and deployment readiness.

NEXT DELIVERABLE

Verified dataset + held-out evaluation + pilot go / no-go

Ubeconn’s next step: convert a working technical foundation into defensible operational evidence.

ubeconn

Sources and scope behind the presentation

Project-local evidence only. No external market, cost-saving or detection-accuracy claims are made.

Platform and deployment	docs/map-app.md; src/Mapui/{api,signals,metrics,live}.py
Asset inventory and mapping	docs/masts.md; docs/channel-mapping.md; results/masts/*.csv
Database snapshot	AGENTS.md; .local/verification.json (16 September 2026)
Discovery pilot	docs/event-discovery.md; results/das-discovery-pilot-20260916/
Supervised baseline	docs/event-ml.md; src/Capture/event_ml.py
Application image	Original .local/screenshots/overview.png; map attribution retained

READING THE STATUS

Implemented / field validated
Inferred / confirmed Proposed / committed

All numerical project claims refer to the documented snapshot or recorded pilot run.

ubeconn

A simple identity, grounded in ube purple

House-report hierarchy and IBM Plex typography, adapted to Ubeconn’s colour profile.

ubeconn

PLAIN WHITE WORDMARK

Deep plum
#1D1429

Ube
#9D6BD0

Lavender
#CFADF0

Soft white
#F6F0FC

IBM Plex Sans Condensed

IBM Plex Sans / Body IBM Plex Mono / Labels

Dark backgrounds, generous spacing, restrained colour and clear evidence labels.